Predicting variable-speed vibration

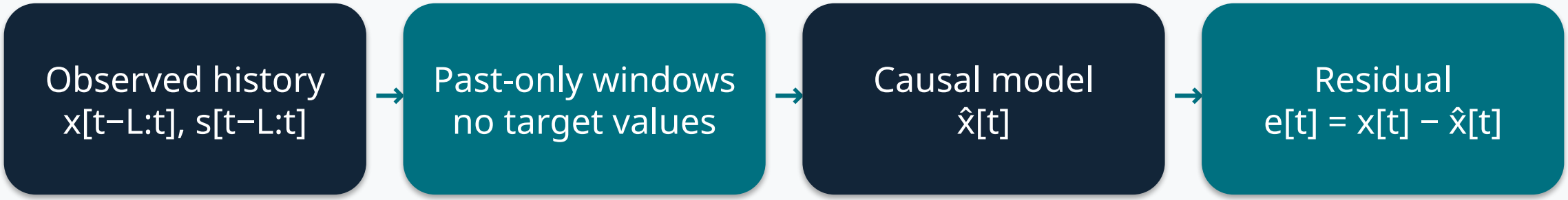
Project progress · 8-minute speaking target · Author / group: add names

Can a compact causal model explain vibration efficiently—and leave interpretable residuals?

- Best measured validation configuration: lpv_L256_s17_a1
- Common-index MSE: 0.3612; independent test unavailable
- 200 separate fault-candidate records processed; physical fault accuracy unverified

A causal prediction task

Previously observed samples predict the next sample; no future vibration



Example: x[64] uses samples 0…63 only. One-step teacher forcing; no x[64] in model inputs.

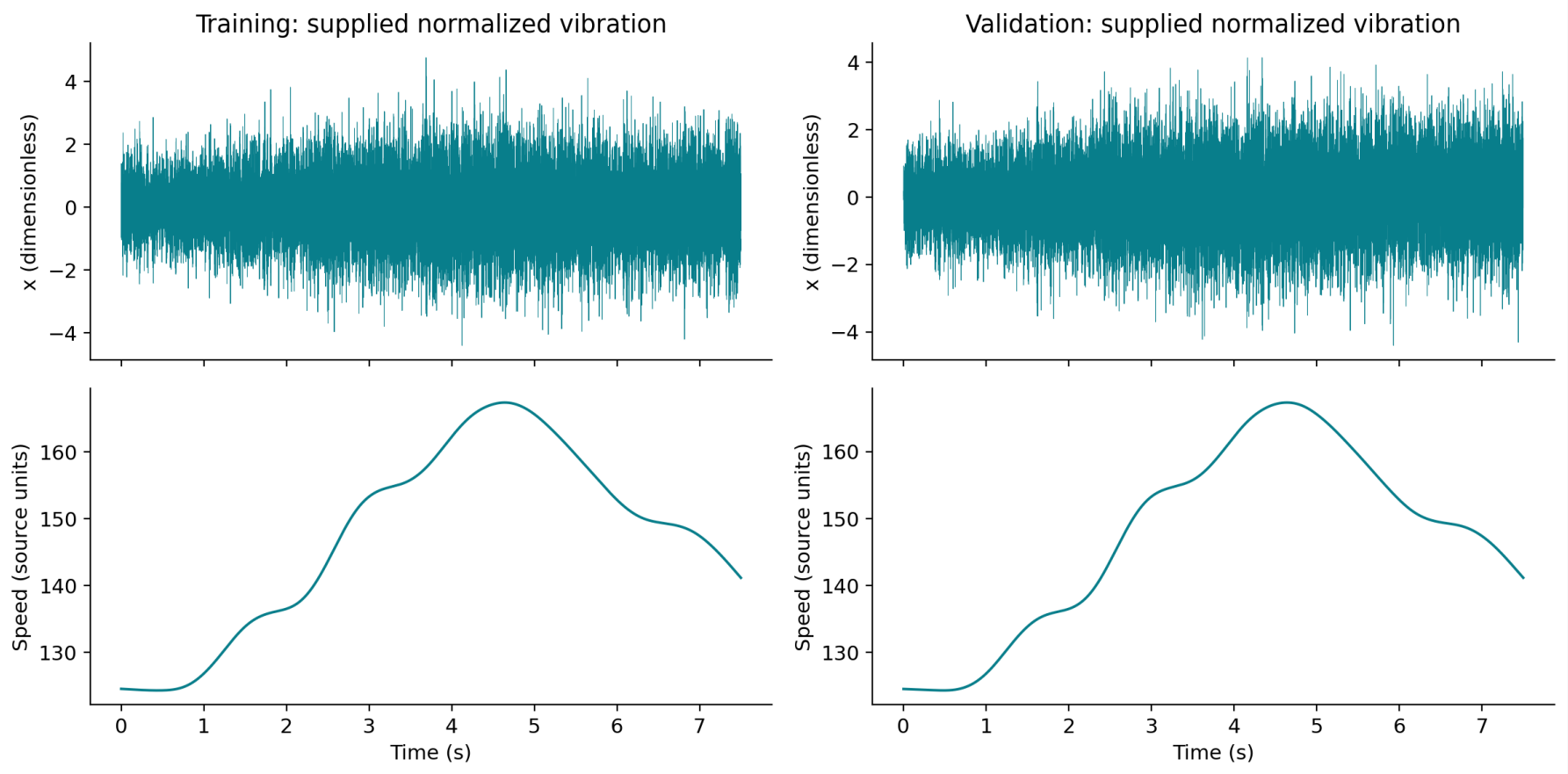
What the accessible data establish

Measured audit; labels and calibration are not invented

- Train: 24,000 samples · Validation: 24,000 samples · fs = 3,200 Hz
- Provided x and spdnorm are used exactly; no record-wise rescaling
- Speed trajectory correlation: 0.999999571
- Speed units / reference health state unverified; normalized x is dimensionless
- Fault candidates: 200 separate 10-second records in 20 MAT parts
- sigtest.mat and sigfault2.mat not supplied

Operating speed changes the signal

Train and validation each contain one 7.5-second recording



Source: data/raw/sigtrain.mat and sigvalid.mat; hashes in data/manifest.json.

A fair evaluation contract

One-step teacher forcing · common burn-in · validation only

- Fit on sigtrain; select epochs/configurations using sigvalid
- Score all models at zero-based indices 256…23,999: 23,744 targets
- Observed past vibration; speed available only through t−1
- No random window split presented as independent generalization
- Validation is reused for selection: final unbiased test performance remains unknown

Strong baselines come first

Least squares, ridge regularization and speed-dependent coefficients

Ridge AR: minimize $\|y - Xb\|^2 + \alpha\|b\|^2$, $\alpha = 1$

- Zero, training-mean and persistence sanity checks
- OLS and ridge AR; ARX appends past speed and speed^2
- LPV-AR: lag coefficients vary polynomially with speed at t−1
- Measured ridge AR(256) validation MSE: 0.420058 (zero: 1.104077)
- LPV-AR surrogate has no ARMA innovation recursion; no exact paper reproduction

Compact nonlinear alternatives

MLP, unidirectional GRU and explicitly causal dilated TCN

MLP
lag vector → scalar

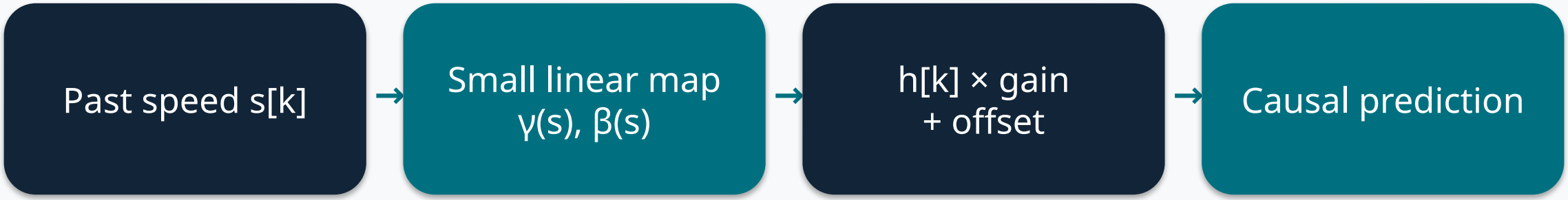
GRU
update / reset gates

TCN
causal dilations

Three separate model alternatives; compact networks, bounded epochs and recorded seeds. TCN: two kernel-3 convolutions per block; receptive field = 1 + 4 Σ dilation.

A falsifiable speed-conditioning proposal

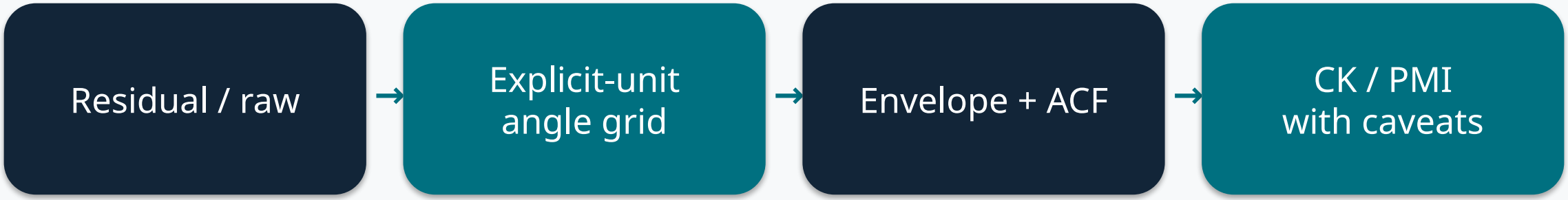
Established FiLM, adapted to observed past speed



Hypothesis: operating-speed information helps beyond extra capacity.
Controls: late concatenation, larger unconditioned TCN and off-line phase-shifted speed (negative control only).

Residuals are diagnostic candidates

Python numerical validation is separate from real fault evidence



Literal MATLAB PMI selects lag T−1; corrected variant selects T. Numerical tests and explicit unit assumptions support descriptive indicators; physical fault accuracy is unverified.

200 separate fault-candidate records processed; physical fault accuracy unverified

Evidence, blockers and next steps

Updated PDF rubric: organization 4 · clarity 6 · understanding 8 · novelty 2%

- Measured best: lpv_L256_s17_a1, validation MSE 0.3612
- LPV-AR beats the tested neural models under this budget
- FiLM speed-TCN mean MSE is 1.90% higher than TCN (3 seeds)
- 200 separate fault-candidate records processed; physical fault accuracy unverified
- Next evidence: independent test and verified physical speed units / health labels